# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Industry Problem Solving Task

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

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**COURSE OUTCOME COVERED**

**CO4:** *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5

Duration :60 Minutes

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**Code of Conduct**

I **B RUTHESH (19251465)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student**:** **B RUTHESH (192521465)**

**Assessment Tasks**

**Industry Problem Solving Task**

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.
Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
   A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain
   how analytics can support decision-making.

**Industry Problem Solving Rubric**

| **Criteria** | **Weightage** | **Level 4 - Exemplary** | **Level 3 - Proficien t** | **Level 2 - Developing** | **Level 1 - Beginning** |
|---|---|---|---|---|---|
| Understanding of Industry Context | 20% | Demonstrates strong understanding of real-world problem context | Good understanding | Basic understanding | Weak understanding |
| Application of Big Data Concepts | 25% | Applies highly relevant big data ideas and tools | Mostly appropriate application | Partial application | Weak application |
| Solution Design | 25% | Solution is practical, scalable, and well-reasoned | Reasonable solution | Basic solution with gaps | Unclear solution |
| Security / Risk Awareness | 15% | Strong awareness of security and operational risks | Reasonable awareness | Limited awareness | Poor awareness |
| Clarity and Professionalism | 15% | Well-structured and professional response | Generally clear | Moderately clear | Poorly presented |

**Total Marks Awarded:**

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## CO4_AT2: Industry Problem Solving Task

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|---|---|
| **Course Name** | Cloud Computing and Big Data Analytics |
| **Course Code** | CSA-1507 |
| **Course Outcome** | CO4: Analyze big data characteristics and challenges across domains including security and application aspects |
| **Assessment** | Assessment Tool 2 - Industry Problem Solving Task |
| **Name** | B RUTHESH |
| **Register No.** | 192521465 |

### Problem 1: Retail Data Optimization

A retail chain with both physical stores and an online platform generates large volumes of sales, inventory, and customer-behavior data, and needs real-time inventory tracking, demand forecasting, and personalised marketing.

### 1. Understanding the Industry Context

Retail operates on thin margins where stock-outs and overstocking directly hurt revenue. Data arrives from three fronts simultaneously: point-of-sale (POS) terminals in stores, the e-commerce platform, and warehouse/inventory systems. The business goal is to unify these streams so that inventory decisions and marketing offers are made in near real time rather than from a next-day batch report.

### 2. Proposed Big Data Architecture

A Lambda-style architecture is well suited here because it needs both a fast real-time path (for inventory and alerts) and a batch path (for deep demand forecasting and trend analysis).

| Layer | Purpose | Representative Tools |
|---|---|---|
| Ingestion | Capture POS transactions, web clickstream, app events and RFID/IoT sensor feeds as they occur | Apache Kafka / AWS Kinesis, Flume |
| Storage | Durable, low-cost storage for raw and processed data | Hadoop HDFS / Amazon S3 (data |

| Layer | Purpose | Representative Tools |
|---|---|---|
| | | lake), Cassandra or DynamoDB for fast key-value lookups |
| Batch Processing | Historical trend analysis, model training, aggregations | Apache Spark / Hive on top of the data lake |
| Stream Processing | Real-time inventory updates, low-stock alerts | Apache Spark Streaming / Apache Flink |
| Serving Layer | Low-latency query layer for dashboards and apps | Apache HBase / Redis (cache) / a data warehouse such as Snowflake or Redshift |
| Analytics & ML | Demand forecasting, recommendation engine | Spark MLlib, TensorFlow, a feature store |
| Visualization | Business dashboards for store and category managers | Power BI / Tableau |

## 3. Analytics Models

- **Demand forecasting:** time-series models (ARIMA, Prophet) or gradient-boosted trees (XGBoost) trained on historical sales, seasonality, promotions and weather data to predict SKU-level demand per store.
- **Real-time inventory tracking:** a streaming job continuously decrements stock counts as POS/online sales events arrive and triggers reorder alerts when a threshold is crossed.
- **Personalised marketing:** collaborative filtering and market-basket analysis (association rule mining, e.g. Apriori/FP-Growth) to recommend products, combined with customer segmentation (k-means clustering) for targeted campaigns.

## 4. Security / Risk Awareness

- Encrypt customer PII (payment details, addresses) at rest and in transit; tokenise card data to remain PCI-DSS compliant.

- Role-based access control so that only authorised analysts can view raw customer-level data; aggregate/anonymised views for wider teams.
- Data-quality checks at ingestion to prevent bad sensor/POS readings from corrupting forecasts.
- A disaster-recovery/replication plan for the streaming pipeline so a broker or node failure does not stop real-time inventory updates.

### 5. Conclusion

The Lambda architecture lets the retailer react to stock and sales events within seconds while still supporting deep, model-driven demand forecasting and personalisation from the same underlying data lake, giving both operational responsiveness and strategic insight.

## Problem 2: Social Media Fraud Detection

A social media platform processes billions of daily interactions (posts, comments, likes, shares) and must detect fake accounts and misinformation in near real time.

### 1. Understanding the Industry Context

At this scale, the challenge is volume, velocity and variety together: text, images, video, network/graph relationships and behavioural signals must all be examined quickly, because misinformation and fake-account networks spread fastest in the first few hours after creation.

### 2. Proposed Big Data Architecture

| Layer | Purpose | Representative Tools |
| --- | --- | --- |
| Ingestion | Capture every post/comment/like/share event as a stream | Apache Kafka (high-throughput event bus) |
| Stream Processing | Score events for fraud/misinformation signals within seconds | Apache Flink / Spark Structured Streaming |
| Storage | Raw event archive and processed feature store | HDFS / S3 data lake, Cassandra/HBase for graph-adjacent lookups |

| Layer | Purpose | Representative Tools |
|---|---|---|
| Graph Analysis | Model relationships between accounts to expose fake-account clusters/bot networks | Apache Spark GraphX / Neo4j |
| ML Serving | Real-time inference for account/content scoring | TensorFlow Serving / a low-latency model-serving layer behind Flink |
| Batch/Retraining | Periodically retrain models on newly labelled data | Spark on the data lake |

### 3. Role of Machine Learning and Stream Processing

- **Stream processing frameworks (Flink/Spark Streaming):** apply windowed aggregation (e.g., events-per-minute per account) so that abnormal burst behaviour, a hallmark of bot activity, is flagged as it happens rather than in an overnight batch.
- **Fake-account detection:** supervised classifiers (random forest / gradient boosting) trained on account-age, posting-frequency, device fingerprint and network features; graph algorithms (community detection, PageRank-style centrality) reveal clusters of accounts that behave in a coordinated, bot-like way.
- **Misinformation detection:** NLP models (transformer-based text classifiers) score content credibility, cross-referenced against a knowledge base of verified facts and prior fact-checked claims; image/video hashing detects re-shared manipulated media.
- **Feedback loop:** human moderator decisions and user reports are fed back as labelled training data so models improve continuously (active learning).

### 4. Security / Risk Awareness

- False positives can silence legitimate users, so a human-in-the-loop review step is required before permanent account action, not fully automated takedown.
- Adversaries adapt quickly, so models need frequent retraining and A/B evaluation to avoid concept drift.

- User data used for behavioural modelling must respect privacy regulations (e.g., GDPR); store only the features needed, not raw personal content, in the ML feature store.
- Rate-limiting and API-abuse detection at the ingestion layer protects the platform itself from being used to mass-create fake accounts.

### 5. Conclusion

Combining a low-latency stream-processing backbone with graph analytics and continuously retrained ML models allows the platform to catch coordinated fake-account and misinformation campaigns within minutes of them starting, while a human review layer keeps false positives in check.

## Problem 3: Government Data Platform

A government agency is building a national data platform integrating census, economic, and geospatial datasets, which multiple departments must access in an interoperable, private and secure way.

### 1. Understanding the Industry Context

Government data platforms differ from commercial ones in priority: interoperability across legacy departmental systems, long-term data stewardship, legal compliance and public accountability matter as much as analytics performance. Datasets are heterogeneous (structured census tables, economic time-series, geospatial shapefiles/GIS layers) and owned by different departments with different access rules.

### 2. Proposed Architecture & Data Management Strategy

| **Component** | **Purpose** | **Representative Tools/Standards** |
|---|---|---|
| Data Lake / Hub | Central repository holding raw datasets from every department in their native formats | HDFS / cloud object storage (S3-compatible), organised into raw / curated / published zones |

| Component | Purpose | Representative Tools/Standards |
| --- | --- | --- |
| Metadata Catalog & Data Governance | Makes datasets discoverable, tracks lineage, ownership and sensitivity classification | Apache Atlas / a data catalog with a formal data-governance policy |
| Master Data Management (MDM) | Resolves identity across departments (e.g., matching a citizen/business record across census and economic datasets) | MDM tooling with defined golden-record rules |
| Interoperability Layer | Common APIs and schemas so departments exchange data without bespoke integrations | REST/GraphQL APIs, open standards (e.g., OGC standards for geospatial data) |
| Geospatial Processing | Handle GIS layers and spatial queries | GeoServer / PostGIS / Hadoop-GIS |
| Access & Analytics | Department-specific dashboards and cross-department analytics | Spark for batch analytics, BI tools for reporting |

## 3. Governance Strategy

- **Data classification:** every dataset is tagged (public, restricted, confidential) and access policy is derived automatically from that tag.
- **Federated ownership, central stewardship:** each department remains the data owner and is accountable for quality, while a central governance office defines shared standards, naming conventions and data-quality rules so integration is possible.
- **Data-sharing agreements:** formal inter-department agreements specify purpose limitation — data shared for one use case cannot silently be repurposed.
- **Audit and lineage tracking:** every access and transformation is logged, so any dataset's provenance can be traced end to end for public accountability.

## 4. Security / Privacy Requirements

- Role-based and attribute-based access control (ABAC) so a department only sees the fields relevant to it (e.g., economic analysts see aggregated, not individually identifiable, census records).
- Anonymisation/pseudonymisation and statistical disclosure control (e.g., k-anonymity) before any citizen-level census data is shared across departments.
- Encryption at rest and in transit, with keys managed by a dedicated key-management service, and network segmentation between departments.
- Regular independent security audits and compliance mapping to national data-protection law.

### 5. Conclusion

A federated-ownership, centrally-governed data hub with strong metadata cataloguing, standard interoperability APIs and strict access/privacy controls lets departments share census, economic and geospatial data safely without each one having to solve integration and security independently.

## Problem 4: Banking Fraud Detection

A bank processes millions of transactions per minute and must detect fraud in real time by analysing both historical and streaming data, under strict low-latency, high-accuracy requirements.

### 1. Understanding the Industry Context

Banking fraud detection is one of the strictest real-time big-data use cases: a decision (approve/decline/flag) must typically be made in well under a second per transaction, false declines damage customer trust and revenue, and false negatives cause direct financial loss and regulatory exposure.

### 2. Proposed Big Data Pipeline

| Layer | Purpose | Representative Tools |
|---|---|---|
| Ingestion | Capture every card-swipe/transfer/online transaction event | Apache Kafka (guarantees ordered, durable delivery at high throughput) |

| Layer | Purpose | Representative Tools |
|---|---|---|
| Real-time Scoring | Score each transaction against a fraud model within milliseconds, in the transaction's critical path | Apache Flink / Spark Structured Streaming with a low-latency model server |
| Feature Store | Maintain rolling features (spend velocity, location deviation, merchant category patterns) updated in real time | Redis / an in-memory feature store |
| Historical Storage | Long-term storage for model training and audits | HDFS / S3 data lake, a relational/columnar warehouse for compliance reporting |
| Batch ML Training | Periodically retrain fraud models on labelled historical fraud cases | Spark MLlib / distributed gradient-boosting frameworks |
| Case Management | Route flagged transactions to fraud analysts | A workflow/queueing system integrated with the bank's core system |

## 3. Technologies and Algorithms

- **Rule-based pre-filters:** fast, explainable checks (transaction above a threshold, impossible travel between two locations) run first to catch obvious fraud cheaply before invoking heavier ML models.
- **Supervised ML models:** gradient-boosted trees (XGBoost/LightGBM) or logistic regression trained on labelled historical fraud/non-fraud transactions, chosen for their balance of accuracy and low inference latency.
- **Anomaly detection:** unsupervised models (isolation forest, autoencoders) catch novel fraud patterns not present in historical labels.
- **Graph-based detection:** account/merchant/device relationship graphs expose fraud rings and mule-account networks that individual-transaction scoring would miss.

- **Streaming windowed aggregates:** velocity checks such as 'more than N transactions in M minutes' computed continuously in the stream layer.

### 4. Security / Risk Awareness

- The pipeline itself is a high-value target, so it must be PCI-DSS compliant: encrypted data, tokenised card numbers, and strict network isolation for the payment-processing path.
- Model explainability (e.g., SHAP values) is required so declined transactions can be justified to customers and regulators.
- High availability and failover for the streaming layer are critical — any downtime in real-time scoring either blocks legitimate transactions or forces a fallback to riskier rule-only mode.
- Continuous model monitoring for drift, since fraud patterns evolve quickly and a stale model degrades both false-positive and false-negative rates.

### 5. Conclusion

A streaming-first pipeline — cheap rule filters, then a low-latency ML/anomaly-detection layer backed by a real-time feature store, with graph analytics for organised fraud — lets the bank meet strict latency requirements without sacrificing detection accuracy, while batch retraining keeps the models current.

## Problem 5: Agriculture Smart Farming

A smart-farming system collects soil, weather and crop data from distributed sensors and drones, and farmers need insights to improve yield and resource usage.

### 1. Understanding the Industry Context

Precision agriculture differs from the other cases in its data profile: connectivity in rural fields is often intermittent, data arrives from many small, low-power distributed sources (soil-moisture sensors, weather stations, drones), and decisions (irrigation, fertiliser application) are made on a daily/seasonal cycle rather than in milliseconds — though some interventions such as frost or pest alerts do need near-real-time response.

## 2. Proposed Big Data Solution

| Layer | Purpose | Representative Tools |
|---|---|---|
| Edge/Field Layer | Collect soil moisture, temperature, humidity and crop-health data locally; buffer data when connectivity drops | IoT sensor nodes with edge gateways, drone-mounted multispectral cameras |
| Ingestion | Transmit sensor/drone data to the cloud once connectivity is available | MQTT (lightweight IoT protocol) feeding into Apache Kafka |
| Storage | Store raw sensor readings, imagery and historical yield records | Hadoop HDFS / a cloud data lake |
| Processing | Clean, aggregate and join multi-source data (soil + weather + imagery) | Apache Spark |
| Analytics/ML | Yield prediction, irrigation scheduling, pest/disease detection from imagery | Spark MLlib for tabular models, CNN-based image models for drone imagery |
| Delivery | Deliver recommendations to farmers, including in low-connectivity areas | A mobile app with offline caching, SMS alerts for critical events |

## 3. How Analytics Support Decision-Making

- **Precision irrigation:** soil-moisture sensor trends combined with weather forecasts feed a regression/rule-hybrid model that recommends exactly how much to irrigate and when, reducing water waste.
- **Yield prediction:** historical yield, soil nutrient levels and weather patterns are used to train models (random forest/regression) that forecast expected yield per field zone, helping with planning and financing.
- **Crop health / pest detection:** drone multispectral imagery is analysed with convolutional neural networks to detect disease or pest stress earlier than the human eye, enabling targeted rather than blanket pesticide use.

- **Resource optimisation:** combining fertiliser-application data with yield outcomes over seasons highlights which zones are over- or under-fertilised (variable-rate application maps).

## 4. Security / Risk Awareness

- Sensor and drone networks are exposed physically, so device authentication and firmware integrity checks are needed to prevent spoofed readings from corrupting recommendations.
- Intermittent connectivity requires robust edge buffering and conflict resolution so no data is silently lost or duplicated when sync resumes.
- Farm-level data (yield, financial performance) is commercially sensitive; access controls should ensure a farmer's data is not exposed to competitors or misused by data aggregators without consent.
- Model recommendations (e.g., irrigation) should have safe fallbacks/human override, since a bad automated decision can damage an entire season's crop.

## 5. Conclusion

An edge-to-cloud pipeline that buffers data at the field level, ingests it via lightweight IoT protocols, and applies both tabular ML (yield, irrigation) and image-based deep learning (crop health) gives farmers timely, resource-efficient decision support even in low-connectivity rural environments.